

Shielding Effectiveness of Stratified Media

Electromagnetic plane waves are the simplest solution of the time-harmonic Maxwell equations in a homogeneous and source-free spatial region. The study of their properties is useful for better understanding the behavior of more complex fields. For instance, the far field radiated by an arbitrary source has, locally and sufficiently far from the source, the characteristics of a plane wave (this usually allows an EM field impinging on a given structure to be approximated as a plane wave). In addition the exact field produced by any source can be expressed in terms of a continuous superposition of elemental plane-wave components (plane-wave spectrum representation). On the other hand, stratified media (in planar, cylindrical, and spherical configurations) are the simplest example of inhomogeneous media and are often considered as models of many shields.

4.1 ELECTROMAGNETIC PLANE WAVES: DEFINITIONS AND PROPERTIES

A plane wave (with a time-harmonic behavior $e^{j\omega t}$, suppressed for the sake of conciseness) is a frequency-domain EM field mathematically expressed as

$$\begin{aligned}\mathbf{E}(x, y, z) &= \mathbf{E}_0 e^{-j(k_x x + k_y y + k_z z)} = \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}}, \\ \mathbf{H}(x, y, z) &= \mathbf{H}_0 e^{-j(k_x x + k_y y + k_z z)} = \mathbf{H}_0 e^{-j\mathbf{k} \cdot \mathbf{r}},\end{aligned}\tag{4.1}$$

where $\mathbf{E}_0$ and $\mathbf{H}_0$ are constant complex vectors, and k_x , k_y , and k_z are complex scalars that define the wavevector $\mathbf{k} = k_x \mathbf{u}_x + k_y \mathbf{u}_y + k_z \mathbf{u}_z$. The vectors $\mathbf{E}_0$ and $\mathbf{H}_0$ define the

polarization of the plane wave, namely the temporal evolution of the vector direction in the plane on which the vector lies. In particular, according to the locus described by the tip of the vector, polarization can be linear, circular, or, in the most general case, elliptic. For circular and elliptic polarizations, the sense of rotation for increasing time can be clockwise or counterclockwise with respect to a fixed direction. The real and (the opposite of) the imaginary part of the wavevector $\mathbf{k}$ are the so-called phase vector $\boldsymbol{\beta}$ and attenuation vector $\boldsymbol{\alpha}$, respectively, so that $\mathbf{k} = \boldsymbol{\beta} - j\boldsymbol{\alpha}$ (note that $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ are real vectors). Alternatively, the propagation vector $\boldsymbol{\gamma} = \boldsymbol{\alpha} + j\boldsymbol{\beta}$ can be used (i.e., $\mathbf{k} = -j\boldsymbol{\gamma}$). In any case, the magnitude of $\boldsymbol{\beta}$ gives the phase shift per unit length along the direction of $\boldsymbol{\beta}$, while the magnitude of $\boldsymbol{\alpha}$ determines the rate of attenuation along the direction of $\boldsymbol{\alpha}$. From Maxwell's equations it follows that the wavevector must satisfy the "separation equation"

$$\mathbf{k} \cdot \mathbf{k} = k_x^2 + k_y^2 + k_z^2 = k^2, \quad (4.2)$$

where k is the *wavenumber* of the medium defined as

$$k = \omega \sqrt{\mu \left(\varepsilon - j \frac{\sigma}{\omega} \right)} = \omega \sqrt{\mu \varepsilon_c} \quad (4.3)$$

(the principal branch of the square root is chosen so that the imaginary part of k is nonpositive). Recall that in the frequency domain the complex electric permittivity ε_c is introduced to account for conductive losses, meaning $\varepsilon_c = \varepsilon - j\sigma/\omega$. From Maxwell's equations, it follows that the wavevector is related to the electric and magnetic field vectors through

$$\begin{aligned} \mathbf{k} \times \mathbf{E} &= \omega \mu \mathbf{H}, \\ \mathbf{k} \times \mathbf{H} &= \omega \varepsilon_c \mathbf{E}. \end{aligned} \quad (4.4)$$

Equations (4.4) imply that $\mathbf{E} \cdot \mathbf{H} = 0$, $\mathbf{k} \cdot \mathbf{E} = 0$, and $\mathbf{k} \cdot \mathbf{H} = 0$. From (4.4) it also follows that $\mathbf{E} \cdot \mathbf{E} = \eta^2 \mathbf{H} \cdot \mathbf{H}$, where η is the *intrinsic impedance* of the medium defined as

$$\eta = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon - j\frac{\sigma}{\omega}}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \quad (4.5)$$

(in this case the principal branch of the square root is chosen so that the real part of η is nonnegative [2]).

A characterization of a plane wave can be obtained by introducing the direction angles (ϕ, θ) . Such direction angles (which are in general complex) are defined as

$$\begin{aligned} k_x &= k \cos \phi \sin \theta, \\ k_y &= k \sin \phi \sin \theta, \\ k_z &= k \cos \theta. \end{aligned} \quad (4.6)$$

Uniform (or *homogeneous*) plane waves are an important subset of plane waves. A uniform plane wave is defined as a plane wave for which both the direction angles (ϕ , θ) are real. Therefore, for a uniform plane wave, the wavevector can be written as $\mathbf{k} = k \mathbf{u}_r$, where $\mathbf{u}_r$ is a real unit vector given by

$$\mathbf{u}_r = \cos \phi \sin \theta \mathbf{u}_x + \sin \phi \sin \theta \mathbf{u}_y + \cos \theta \mathbf{u}_z. \quad (4.7)$$

For a uniform plane wave, the phase and attenuation vectors point in the same direction of the unit vector $\mathbf{u}_r$. It can easily be shown that inside an isotropic medium this is also the direction of the time-averaged power-flow vector (i.e., the real part of the Poynting vector $\mathbf{\Pi}$). Therefore, for a uniform plane wave, all the vectors $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{\Pi}$ point in the same direction: such a direction can be uniquely defined as the *direction of propagation* of the uniform plane wave. Since $\mathbf{u}_r \cdot \mathbf{E} = \mathbf{u}_r \cdot \mathbf{H} = 0$ (i.e., $E_r = H_r = 0$), a uniform plane wave is said to be transverse electromagnetic (or TEM) to the direction of propagation $\mathbf{u}_r$; furthermore we have $|\mathbf{E}| = |\eta| |\mathbf{H}|$.

It is worth noting that for lossless media the wavenumber k is real, so the attenuation vector is zero. Moreover it is important to note that since for non-uniform plane waves the direction angles are complex, their attenuation vector is never zero and cannot be parallel to the phase vector, so a more involved relation exists among the $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{\Pi}$ vectors. Therefore the direction of propagation cannot be uniquely defined for non-uniform plane waves.

As is well known [1], an EM field in a source-free homogeneous region can be represented as the sum of two types of fields, a field that is transverse magnetic to an arbitrary direction x (TM $_x$, for which $H_x = 0$) and a field that is transverse electric to x (TE $_x$, for which $E_x = 0$). A general plane wave can thus be written as the sum of a TM $_x$ plane wave and a TE $_x$ plane wave. By defining the plane of incidence as the plane containing the phase vector $\boldsymbol{\beta}$ and the unit vector $\mathbf{u}_x$, the TM $_x$ plane wave has the electric field lying in the plane of incidence (and it is also called *E* wave, vertically polarized wave, or *p*-polarized wave), while the TE $_x$ plane wave has the magnetic field lying in the plane of incidence (and it is also called *H* wave, horizontally polarized wave, or *s*-polarized wave). Actual sources generally give rise to EM fields that are not uniform TEM (or TE, TM) plane waves. However, as already mentioned above, at distances far enough from the sources (i.e., for distances much larger than the longest significant wavelength associated with the frequency spectrum of the source and than its largest dimension), the assumption of a uniform plane wave is often a reasonable approximation.

In the following, we will first consider the problem of plane-wave incidence on a planar shield. The convenient formalism of the so-called equivalent transmission line will next be introduced. Such a formalism has the advantage of automatically including the boundary conditions for the fields. Last, the analysis of shielding in the presence of curved surfaces and near-field sources will be discussed.

4.2 UNIFORM PLANE WAVES INCIDENT ON A PLANAR SHIELD

The simplest shielding configuration is that depicted in Figure 4.1 where a uniform plane wave is impinging with an angle θ^{inc} on a single planar shield of infinite extension in the transverse y and z directions (i.e., whose dimensions are large enough to neglect edge effects) with a finite thickness d in the x direction. Although this configuration is almost not realizable and far from usual actual conditions, some basic ideas on shielding are based on this problem. For this reason it has been kept as a starting point of any further and deeper analysis.

The most useful way to solve the problem is based on the analogy existing between uniform plane-wave propagation in a stratified medium and the propagation of voltages and currents in a uniform transmission line. In the next subsection we briefly summarize the key steps in proving such analogy.

4.2.1 Transmission-Line Approach

We thus start with a plane wave propagating in a medium that is linear, homogeneous, stationary, isotropic, and possibly lossy and dispersive. Linearity and stationarity of the constitutive relations characterizing the EM behavior of the shield material guarantee that the boundary value problem may be conveniently posed in the frequency domain, by introducing the magnetic permeability μ and electric permittivity ϵ_c that, under the assumptions above, are complex frequency-dependent scalar quantities, meaning $\mu = \mu' - j\mu''$ (with $\mu' > 0$ and $\mu'' > 0$) and $\epsilon_c = \epsilon'_c - j\epsilon''_c$ (with $\epsilon'_c > 0$ and $\epsilon''_c > 0$). The possible occurrence of negative values of ϵ'_c and/or μ' will be discussed in Chapter 12. It should be noted that the hypothesis of linearity is often fulfilled in consideration of the reduced values of the incident high-frequency fields, while the stationarity assumption is often reasonable because of the different order of magnitude existing between the time scales of EM phenomena and typical changes (e.g., for thermal or corrosion issues) occurring in the materials characteristics.

As was mentioned above, a general plane wave can be written as the sum of a TM_x and a TE_x plane wave. The TM_x propagation in the i th region of a stratified medium

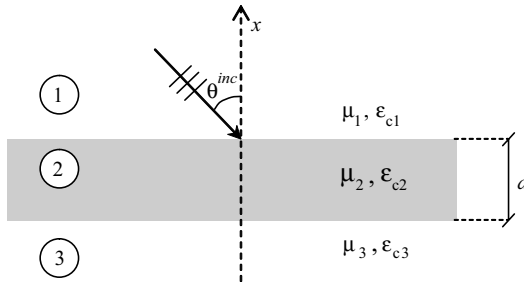


FIGURE 4.1 Uniform plane wave impinging on a shield of finite thickness d and of infinite transverse dimensions.

characterized by its relative permittivity ε_{cri} and permeability μ_{ri} is considered first. In such a case the magnetic field of the plane wave can be expressed as

$$\mathbf{H}^{\text{TM}}(x, y, z) = (\mathbf{u}_y H_y^{\text{TM}} + \mathbf{u}_z H_z^{\text{TM}}) e^{-j(\pm k_{xi}x + k_{yi}y + k_{zi}z)}, \quad (4.8)$$

where $k_{xi}^2 + k_{yi}^2 + k_{zi}^2 = k_i^2 = \omega^2 \mu_i \varepsilon_{ci} = \omega^2 \mu_0 \varepsilon_0 \mu_{\text{ri}} \varepsilon_{\text{cri}} = k_0^2 \mu_{\text{ri}} \varepsilon_{\text{cri}}$. In equation (4.8) the plus sign is chosen for plane waves propagating or decaying in the positive x direction, while the negative sign is for propagation or decay in the minus x direction.

From Maxwell's equations, we can easily derive the expression of the electric field. In particular, for the components of the electric field normal to the x direction,

$$\begin{aligned} E_y(x, y, z) &= \frac{1}{j\omega\varepsilon_{ci}} \mathbf{u}_y \cdot \nabla \times \mathbf{H}(x, y, z) = \pm \frac{k_{xi}}{\omega\varepsilon_{ci}} H_z(x, y, z), \\ E_z(x, y, z) &= \frac{1}{j\omega\varepsilon_{ci}} \mathbf{u}_z \cdot \nabla \times \mathbf{H}(x, y, z) = \mp \frac{k_{xi}}{\omega\varepsilon_{ci}} H_y(x, y, z). \end{aligned} \quad (4.9)$$

These expressions can be recast in the form

$$\mathbf{E}_\tau^{\text{TM}}(x, y, z) = \mp Z_i^{\text{TM}} [\mathbf{u}_x \times \mathbf{H}_\tau^{\text{TM}}(x, y, z)], \quad (4.10)$$

where the subscript τ indicates the vector component transverse to the x direction and the TM_x characteristic impedance Z_i^{TM} is defined as

$$Z_i^{\text{TM}} = \frac{k_{xi}}{\omega\varepsilon_{ci}}. \quad (4.11)$$

With similar steps it can be shown that the transverse components of the TE_x fields are related by

$$\mathbf{E}_\tau^{\text{TE}}(x, y, z) = \mp Z_i^{\text{TE}} [\mathbf{u}_x \times \mathbf{H}_\tau^{\text{TE}}(x, y, z)], \quad (4.12)$$

where the TE_x characteristic impedance Z_i^{TE} is defined as

$$Z_i^{\text{TE}} = \frac{\omega\mu_i}{k_{xi}}. \quad (4.13)$$

It can immediately be seen that the transverse electric and magnetic fields in (4.10) and (4.12) correspond to the voltage and current, respectively, of an equivalent transmission line (TL) with propagation constant k_{xi} and characteristic impedance $Z_i^{\text{TM/TE}}$. In particular, the correspondence is such that

$$\begin{aligned} \mathbf{E}_\tau^{\text{TM/TE}}(x, y, z) &= V^{\text{TM/TE}}(x) e^{-j(k_{yi}y + k_{zi}z)} \mathbf{u}_E, \\ \mathbf{H}_\tau^{\text{TM/TE}}(x, y, z) &= I^{\text{TM/TE}}(x) e^{-j(k_{yi}y + k_{zi}z)} \mathbf{u}_H \end{aligned} \quad (4.14)$$

(where the relation $\mathbf{u}_E \times \mathbf{u}_H = \pm \mathbf{u}_x$ occurs among the unit vectors for propagation in the $\pm x$ direction). In addition, starting from Maxwell's equations, it can be shown that the functions $V^{\text{TM/TE}}$ and $I^{\text{TM/TE}}$ are solutions of TL-like equations. When dealing with planar stratified media (i.e., in which material discontinuities occur only along the x axis) the considerations above are still valid inside each layer.

In the case of a discontinuity between two half-spaces (region 1 and 2), the fulfilment of the boundary conditions at the interface (i.e., continuity of the transverse component of the fields) implies the presence of a reflected wave (in the same half-space where the incident plane wave originates) and a transmitted wave (in the other half-space). The continuity conditions also imply that the transverse dependence of the incident, reflected, and transmitted waves must be the same (i.e., $k_{y1} = k_{y2}$ and $k_{z1} = k_{z2}$). The fact that the transverse wavenumbers are the same in all regions leads to the well-known *law of reflection*, by which the direction angles of the incident and reflected waves are equal, (i.e., $\theta^{\text{inc}} = \theta^{\text{r}}$) and also to *Snell's law*, by which the direction angles inside each of the regions are related to each other:

$$k_1 \sin \theta^{\text{inc}} = k_2 \sin \theta^{\text{t}}. \quad (4.15)$$

In addition, from (4.14), the continuity of the transverse components of the field at the material interfaces implies the continuity of voltage and current at the junction between the two semi-infinite TLs. In general, when a number of layers is considered, voltages and currents are continuous at each material interface. In Figure 4.2 the analogy between a plane-wave transmission problem and propagation along the equivalent TL is sketched. In the figure, the incident, reflected, and transmitted wavevectors are assumed to be real (as the corresponding direction angles, i.e., uniform plane waves are assumed), but the analysis is general and valid also for non-uniform plane waves.

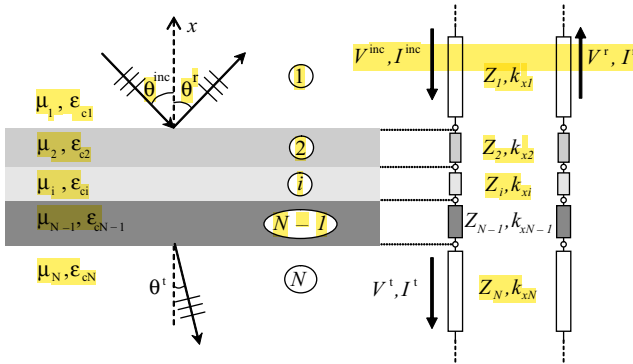


FIGURE 4.2 Uniform plane wave impinging on a planar multilayer shield and equivalent TL model.

It is possible to express the TM_x and TE_x characteristic impedances of the i th layer of a stratified medium in terms of the intrinsic impedance of the medium. In particular, this results in

$$Z_i^{\text{TM}} = \eta_i \cos \theta_i \quad \text{and} \quad Z_i^{\text{TE}} = \frac{\eta_i}{\sin \theta_i} \quad (4.16)$$

or

$$Z_i^{\text{TM}} = \eta_i \sqrt{1 - \left(\frac{k_1}{k_i} \sin \theta_1\right)^2} \quad \text{and} \quad Z_i^{\text{TE}} = \frac{\eta_i}{\sqrt{1 - \left(\frac{k_1}{k_i} \sin \theta_1\right)^2}}, \quad (4.17)$$

where θ_1 is the direction angle of the incident wave in the first region (i.e., the incidence angle).

One might ask why, among all the possible choices, the TM_x - TE_x decomposition has been adopted. As stated above, the functions $V^{\text{TM/TE}}$ and $I^{\text{TM/TE}}$ behave as voltage and current on different TLs, with characteristic impedances $Z_i^{\text{TM/TE}}$. Therefore any plane-wave reflection and transmission problem reduces to a TL problem, giving the exact solution that satisfies all the boundary conditions. One consequence is that TM_x and TE_x plane waves *do not couple* at a boundary. Therefore, if an incident plane wave is expressed as a combination of both TM_x and TE_x waves, the two problems are solved separately, and the solutions are eventually summed to obtain the *total* reflected or transmitted field.

4.2.2 The Single Planar Shield

In general, the two media surrounding the single planar shield of thickness d can be different. In what follows we assume that each material occupying the region i ($i = 1, 2, 3$) is characterized by its relative permittivity ϵ_{cri} and permeability μ_{ri} , although in practical configurations media 1 and 3 are equal (usually air, i.e., $\epsilon_{\text{cr1}} = \epsilon_{\text{cr3}} = \mu_{\text{r1}} = \mu_{\text{r3}} = 1$). In the latter, for uniform plane-wave incidence, the electric and magnetic shielding effectivenesses SE_E and SE_H coincide. Both the incident and the transmitted waves in medium 3 are in fact uniform plane waves, for which $|\mathbf{E}| = |\eta| |\mathbf{H}|$ (so that the ratio of the magnitudes of the electric and magnetic fields is the same in the presence and in the absence of the screen). In particular, with reference to Figure 4.1, the SE of the screen (in dB) is defined as

$$\text{SE}_E = \text{SE}_H = \text{SE} = 20 \log \frac{|\mathbf{E}_3^0|}{|\mathbf{E}_3^S|}, \quad (4.18)$$

where $\mathbf{E}_3$ indicates the electric field in the region 3, while the superscripts 0 and S indicate the absence and the presence of the shield, respectively.

For TE_x -polarized plane waves, it is immediate to see that the ratio between the amplitudes of the total electric fields occurring in (4.18) coincides with the ratio of the amplitudes of the relevant tangential components (no normal component of the

electric field is present in this TE_x case). On the other hand, for TM_x -polarized waves, the amplitude of the normal component of the electric field is proportional to the amplitude of the transverse component through the factor $\tan \theta_3$, where θ_3 indicates the angle of transmission in medium 3. However, such an angle is independent of the presence of the screen, so also in this TM_x case we have

$$\frac{|\mathbf{E}_3^0|}{|\mathbf{E}_3^S|} = \frac{|\mathbf{E}_{3\tau}^0(1 + \tan^2 \theta_3)|}{|\mathbf{E}_{3\tau}^S(1 + \tan^2 \theta_3)|} = \frac{|\mathbf{E}_{3\tau}^0|}{|\mathbf{E}_{3\tau}^S|} \quad (4.19)$$

meaning the ratio of the amplitudes of the total electric fields is equal to the ratio of the amplitudes of the relevant transverse components.

Now, by means of a straightforward TL analysis based on the analogy illustrated in the previous subsection, it is simple to show that for a TM_x or TE_x uniform plane wave impinging on the shield with an angle θ^{inc} , the result is

$$\frac{|\mathbf{E}_3^S|}{|\mathbf{E}_3^0|} = \left| \frac{4Z_2Z_3}{(Z_1 + Z_2)(Z_2 + Z_3)e^{+jk_{x2}d} + (Z_1 - Z_2)(Z_2 - Z_3)e^{-jk_{x2}d}} \right|, \quad (4.20)$$

where $k_{xi} = k_0 \sqrt{\mu_{ri}\varepsilon_{cri} - \mu_{r1}\varepsilon_{cr1} \sin^2 \theta^{\text{inc}}}$ and Z_i ($i = 1, 2, 3$) are the TM_x or TE_x characteristic impedances.

When region 1 and 3 are air, by letting $Z_1 = Z_3 = Z_0$, $Z_2 = Z_S$, and $k_{x2} = k_{xS}$, we reduce (4.20) to

$$\frac{|\mathbf{E}_3^S|}{|\mathbf{E}_3^0|} = \left| \frac{4Z_0Z_S}{(Z_0 + Z_S)^2} e^{-jk_{xS}d} \frac{(Z_0 + Z_S)^2}{(Z_0 + Z_S)^2 - (Z_0 - Z_S)^2 e^{-j2k_{xS}d}} \right|. \quad (4.21)$$

Thus the SE is readily obtained as

$$\text{SE} = -20 \log \left| \frac{4Z_0Z_S}{(Z_0 + Z_S)^2} e^{-jk_{xS}d} \frac{(Z_0 + Z_S)^2}{(Z_0 + Z_S)^2 - (Z_0 - Z_S)^2 e^{-j2k_{xS}d}} \right| = R + A + M, \quad (4.22)$$

where

$$R = 20 \log \left| \frac{(Z_0 + Z_S)^2}{4Z_0Z_S} \right| = 20 \log \left| \frac{(\zeta + 1)^2}{4\zeta} \right|, \quad (4.23a)$$

$$A = 20 \log |e^{jk_{xS}d}| = 8.686 \alpha_{xS}d, \quad (4.23b)$$

$$\begin{aligned} M &= 20 \log \left| \frac{(Z_0 + Z_S)^2 - (Z_0 - Z_S)^2 e^{-j2k_{xS}d}}{(Z_0 + Z_S)^2} \right| \\ &= 20 \log \left| 1 - \frac{(\zeta - 1)^2}{(\zeta + 1)^2} e^{-j2k_{xS}d} \right|, \end{aligned} \quad (4.23c)$$

and $\zeta = Z_0/Z_S$.

The first term, R , in equation (4.22) depends only on the free-space impedance Z_0 and the impedance of the shield medium Z_S , and it accounts for the first field reflection at the two shield interfaces that is due to the mismatch between the two impedances at both the interfaces. R is called the *reflection-loss* term, and it is always positive or null.

The second term, A , is a function of the shield characteristics only, and it accounts for the attenuation that a plane wave undergoes in traveling through an electrical depth equal to $k_x d/k_0$ in the shield material. A is called the *absorption-loss* term, and it is always positive.

The last term, M , is associated with the wave that undergoes multiple reflections and consequent attenuation before passing through the shield. M is called the *multiple-reflection-loss* term, and it may be positive, null, or negative. M is often negligible with respect to the other two terms, especially in the high-frequency range.

To illustrate how the TL approach behaves, the typical values of the intrinsic impedance $\eta = Z^{\text{TE}}(k_x = k) = Z^{\text{TM}}(k_x = k)$ of shielding materials can be compared with the free-space impedance. Three typical shielding materials are selected: a copper casting alloy having relative permeability $\mu_r = 1$ and conductivity $\sigma = 11.8 \cdot 10^6$ S/m, a Duranickel with $\mu_r = 10.58$ (considered independent of frequency) and $\sigma = 2.3 \cdot 10^6$ S/m, and a commercial stainless steel having $\mu_r = 95$ and $\sigma = 1.3 \cdot 10^6$ S/m. The behavior of the intrinsic impedance of these media is plotted in Figure 4.3 in relation to their frequency. It is clearly evident that the intrinsic impedance of these shielding materials is much smaller than the free space impedance which is $\eta_0 \simeq 377 \Omega$. In Figure 4.4, the frequency-dependences of the reflection loss R and of the absorption loss A terms are reported for the same materials as in Figure 4.3, for a planar shield that is 0.25 mm thick under normal (TEM_x) plane-wave incidence. In Figure 4.5, the trend of the multiple-reflection loss term M is also shown; it can be observed that M becomes negligible in comparison with the other two terms

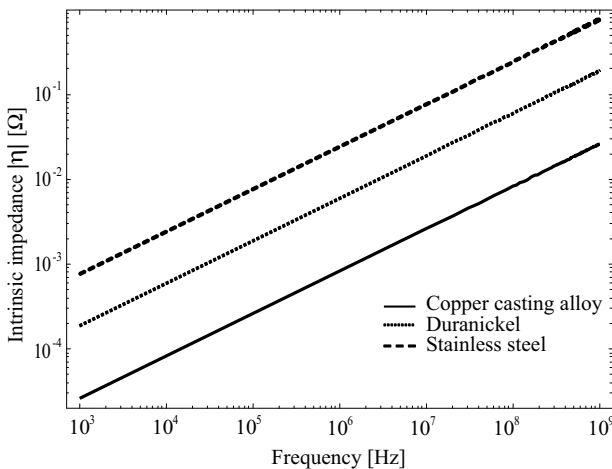


FIGURE 4.3 Intrinsic impedance $|\eta|$ of three typical shielding materials.

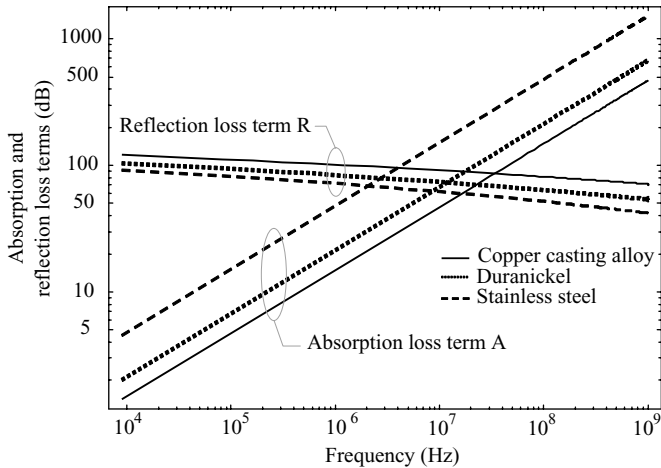


FIGURE 4.4 Reflection-loss R and absorption-loss A terms as functions of frequency for the materials considered in Figure 4.3 and for a shield thickness $d = 0.25$ mm under normal (TEM_x) plane-wave incidence.

above a frequency that depends on the material characteristics and on the shield thickness and that is generally lower than 1 MHz for typical shielding materials.

It is obvious that shielding effectiveness levels in excess of 120 dB are generally not achievable and also not measurable without extraordinary efforts. The preceding figures represent the trends of R and A , and they should be interpreted in the sense that above certain frequencies such quantities exceed any reasonable threshold of shielding level.

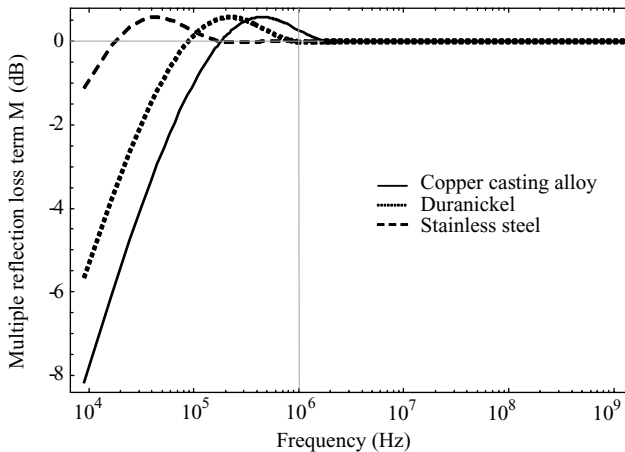


FIGURE 4.5 Multiple-reflection-loss term M as a function of frequency for the materials considered in Figure 4.3 and for a shield thickness $d = 0.25$ mm under normal (TEM_x) plane-wave incidence.

For the common case of sufficiently thick metals, the contribution of the multiple-reflection-loss term M can usually be neglected because of the high penetration loss inside the shielding material. In fact, we can assume $|\zeta| \gg 1$, and rewrite (4.23c) as

$$M = 20 \log \left| 1 - \frac{(\zeta - 1)^2}{(\zeta + 1)^2} e^{-j2k_{xs}d} \right| \cong 20 \log |1 - e^{-2d(1+j)/\delta}|, \quad (4.24)$$

where the frequency-dependent skin depth δ , defined as

$$\delta = \sqrt{\frac{2}{\omega \mu_0 \mu_r \sigma}} \quad (4.25)$$

has been introduced. From (4.25) it is evident that M is almost zero when $d/\delta \gg 1$.

It should be noted that when lossy media are considered in the problem of plane-wave transmission, if $\theta^{\text{inc}} \neq 0$ the transmitted plane wave is necessarily non-uniform. Equation (4.15) is still valid, but the direction angle θ^t assumes complex values. This is also what happens in the well-known case of *total reflection* at the interface between two lossless media. In the latter case, medium 1 (from which the incident wave comes) is denser than medium 2 (i.e., $k_1 > k_2$). The incidence angle θ^{inc} is greater than the *critical angle* $\theta_c = \sin^{-1} \sqrt{k_2/k_1}$, and it can immediately be seen that the wavenumber k_{x2} in medium 2 is purely imaginary. This gives rise to a non-uniform transmitted wave whose fields exponentially decay in medium 2 and whose power flow is in the transverse direction only. The case where three-layered regions are considered is still different. For instance, consider a plane wave propagating in medium 1 with $k_1 > k_0$ and impinging on a planar air gap (region 2) that separates two regions filled with medium 1 (regions 1 and 3). If the incidence angle in region 1 is greater than the critical angle θ_c , the plane waves in the air gap will be non-uniform and, in particular, exponentially decaying (evanescent or reactive fields). However, although individually the backward and forward waves are evanescent, their sum is a standing wave with a power flow in the x direction different from zero. A uniform transmitted wave in fact exists in region 3 (*electromagnetic tunneling*).

Now consider the problem of oblique incidence (for a uniform plane wave) at the interface with an half-space made of a lossy material. Again, the transmitted wave is a non-uniform plane wave. However, it can easily be shown that the attenuation vector $\boldsymbol{\alpha}$ is directed as $-\mathbf{u}_x$, while, if sufficiently high values of the conductivity σ are considered, the angle θ_β between $-\mathbf{u}_x$ and the phase vector $\boldsymbol{\beta}$ is very small, namely $\theta_\beta \simeq 0$. Therefore the transmitted wave can be approximately considered as a uniform plane wave propagating in a direction that is almost normal to the interface, independently of the angle of incidence θ^{inc} . This is also what usually happens in metallic shields. The propagation inside the shield can be approximately considered in a direction that is almost normal to the shield plane in the whole frequency spectrum of interest for shielding applications [4]. Moreover the EM field transmitted

through the shield and propagating in region 3 (always considered as having the same electrical properties as region 1) will have the same direction of propagation it would have in the absence of the shield itself, which is that of the incident field. This will not be true in dealing with anisotropic materials, that is when the permittivity or the permeability of the shield material is expressed by means of a dyadic.

4.2.3 Multiple (or Laminated) Shields

Multiple or laminated shields are obtained by a stratification of two or more sheets of different materials. They are used because of the advantages of shielding performance or for other technical reasons occurring, for instance, when a shielded component is placed into a metallic enclosure. Between adjacent sheets a dielectric (air) gap may or may be not present; such a gap can considerably affect the overall performance of the shield. The simplest configuration consisting of a pair of planar shields separated by an air gap is usually called a double shield [6] in the shielding community, as will be discussed later.

The equivalent circuit of a multiple shield is that already reported in Figure 4.2. The analysis of that configuration is straightforward and can be simply carried out by means of the TL analogy. However, it may be useful to make explicit the equivalence between each shielding layer and a two-port network for which

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix} = \underline{\mathbf{T}} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}. \quad (4.26)$$

where $\underline{\mathbf{T}}$ is the transmission matrix. In particular, for a stratified planar structure made of N layers it results

$$\begin{bmatrix} V_{i-1} \\ I_{i-1} \end{bmatrix} = \begin{bmatrix} \cos(k_{xi}d_i) & -jZ_i \sin(k_{xi}d_i) \\ -\frac{j}{Z_i} \sin(k_{xi}d_i) & \cos(k_{xi}d_i) \end{bmatrix} \begin{bmatrix} V_i \\ I_i \end{bmatrix} = \underline{\mathbf{T}}_i \begin{bmatrix} V_i \\ I_i \end{bmatrix}, \quad (4.27)$$

where d_i is the thickness of the i -th layer, while V_i and I_i are the equivalent voltage and current (corresponding to the transverse electric and magnetic field, respectively) at the interface between the i -th and the $i + 1$ -th layer. Thus, the input–output relationship reads

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \prod_{i=2,3,\dots,N-1} \underline{\mathbf{T}}_i \begin{bmatrix} V_{N-1} \\ I_{N-1} \end{bmatrix} \quad (4.28)$$

to be completed with the two boundary conditions at the first and last interfaces. Using the input–output relationship of the transmission coefficient, and considering the practical case for which region 1 and N are air (i.e., $Z_1 = Z_N = Z_0$), we can express the SE of a multilayer shield as

$$\text{SE} = 20 \log \left| \frac{1}{p} \prod_{n=2}^{N-1} e^{jk_{xn}d_n} (1 - q_n e^{-j2k_{xn}d_n}) \right| \quad (4.29)$$

where

Handwritten diagram showing a sequence of impedances Z_{i-1} , Z_i , and Z_{eq} connected in series, with a separate Z_0 and Z_1 also shown.

$$p = \frac{2Z_0 \prod_{i=2}^{N-1} 2Z_i}{(Z_0 + Z_2)(Z_{N-1} + Z_0) \prod_{i=2}^{N-2} (Z_i + Z_{i+1})}, \quad (4.30a)$$

$$q_i = \frac{(Z_i - Z_{i-1})[Z_i - Z(d_i)]}{(Z_i + Z_{i-1})[Z_i + Z(d_i)]}, \quad (4.30b)$$

$$Z(d_{i-1}) = Z_i \frac{Z(d_i) \cos(k_{xi}d_i) + jZ_i \sin(k_{xi}d_i)}{Z_i \cos(k_{xi}d_i) + jZ(d_i) \sin(k_{xi}d_i)}, \quad (4.30c)$$

$$Z(d_{N-1}) = Z_0.$$

In the case of a planar multilayered shield, the TL formalism allows one to easily determine under which conditions maximum transmission (i.e., absence of reflection) occurs. For instance, in considering a planar shield (region 2) placed between two half-spaces filled with different materials (regions 1 and 3), it is simple to show that when the parameters of the media are chosen so that $Z_2 = \sqrt{Z_1 Z_3}$, the EM energy is totally transmitted from one half-space to the other; the role of the second layer is right that of a quarter-wave transformer matching two lengths of TLs [2]. In addition to the classical work of Schulz et al. [3], some considerations about shielding of planar multilayer structures are presented in [4] and [5].

The explicit SE expression for a laminated shield of two different materials ($N = 4$) can be found in [3]. In [3] the SE expression for a double shield is also reported; the double shield is a practical case of considerable importance constituted by two sheets separated by an air gap ($N = 5$, $Z_1 = Z_3 = Z_N = Z_0$). When the two metallic sheets consist of the same material (i.e., $Z_2 = Z_4$) and of the same thickness (i.e., $d_2 = d_4$), it can be shown that the absorption- and reflection-loss terms are exactly twice those of a single sheet, while the multiple-reflection-loss term is more than twice that of a single sheet; because of the two new interior interfaces between air and the metallic material, the multiple-reflection shielding mechanism is exalted. Figure 4.6 shows the SE of double shields of copper casting alloy ($d_2 = d_4 = \Delta = 0.25$ mm) with $d_3 = \Delta$ and $d_3 = 10\Delta$, compared with the SE of a single layer shield with a double thickness ($d_2 = 2\Delta = 0.5$ mm). It can be seen that the larger the air gap, the more effective is the screen; this is true when the shield is electrically thin. When the shield is electrically thick, the double shield is considerably less effective than the single shield with a double thickness, except at the shielding inter-space resonances. More details on the subject can be found in [3].

4.3 PLANE WAVES NORMALLY INCIDENT ON CYLINDRICAL SHIELDING SURFACES

The EM problem is depicted schematically in Figure 4.7, where a uniform plane wave impinges on an infinitely long cylindrical shield, normally with respect to its

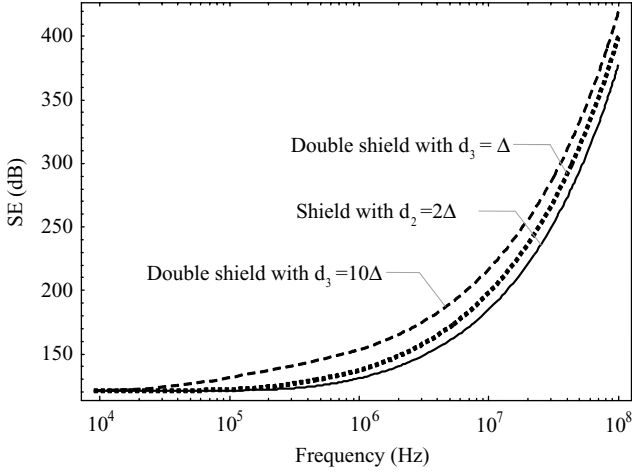


FIGURE 4.6 Comparison between the SE of double shields and a single-layer shield.

axis (i.e., the z axis). The cylindrical shield is characterized by its inner and outer radius ρ_1 and ρ_2 , respectively, and by its constitutive parameters ε_c and μ (in particular, the material is assumed linear, stationary, isotropic, and homogeneous).

The most convenient decomposition for the incident uniform plane wave is in the linearly polarized TM_z and TE_z uniform plane waves. In the case of normal incidence (i.e., $k_z^{\text{inc}} = 0$), the TM_z and TE_z waves in fact do not couple, although in the more general case of oblique incidence (i.e., $k_z^{\text{inc}} \neq 0$) they do. The three regions corresponding to the interior of the shield, to the shield itself, and to the outer space are denoted as region 1, 2, and 3, respectively:

$$\begin{aligned} \rho < \rho_1 & \quad (\text{Region 1}), \\ \rho_1 \leq \rho < \rho_2 & \quad (\text{Region 2}), \\ \rho_2 \leq \rho & \quad (\text{Region 3}). \end{aligned}$$

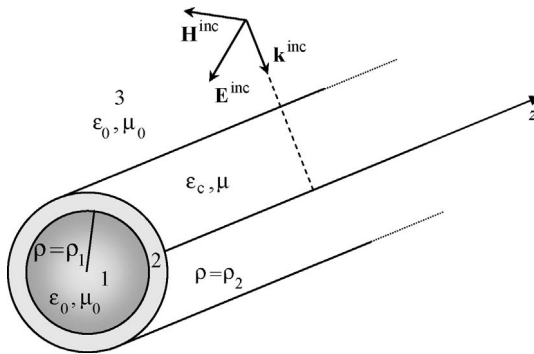


FIGURE 4.7 Uniform plane wave incident on an infinitely long cylindrical shield, normally with respect to its axis.

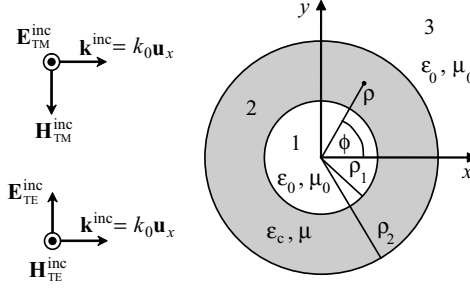


FIGURE 4.8 TM_z and TE_z uniform plane waves normally incident on an infinitely long cylindrical shield.

Since both the incident field and the structure are invariant along z , the problem is essentially a two-dimensional (2D) problem. Let us first consider a uniform TM_z -polarized plane wave normally incident on the shield, as illustrated in Figure 4.8.

Without loss of generality, we can assume that the wave propagates along the x axis of a Cartesian coordinate system having the z axis along the cylindrical shield axis. The electric and magnetic fields are expressed as

$$\begin{aligned} \mathbf{E}^{\text{inc}}(x, y, z) &= E_z^{\text{inc}}(x) \mathbf{u}_z = E_0^{\text{inc}} e^{-jk_0 x} \mathbf{u}_z, \\ \mathbf{H}^{\text{inc}}(x, y, z) &= H_y^{\text{inc}}(x) \mathbf{u}_y = H_0^{\text{inc}} e^{-jk_0 x} \mathbf{u}_y = -\frac{E_0^{\text{inc}}}{\eta_0} e^{-jk_0 x} \mathbf{u}_y, \end{aligned} \quad (4.31)$$

or, in cylindrical coordinates, as

$$\begin{aligned} \mathbf{E}^{\text{inc}}(\rho, \phi, z) &= E_z^{\text{inc}}(\rho, \phi) \mathbf{u}_z = E_0^{\text{inc}} e^{-jk_0 \rho \cos \phi} \mathbf{u}_z, \\ \mathbf{H}^{\text{inc}}(\rho, \phi, z) &= H_\rho^{\text{inc}}(\rho, \phi) \mathbf{u}_\rho + H_\phi^{\text{inc}}(\rho, \phi) \mathbf{u}_\phi = -\frac{E_0^{\text{inc}}}{\eta_0} (\mathbf{u}_\rho \sin \phi + \mathbf{u}_\phi \cos \phi) e^{-jk_0 \rho \cos \phi}. \end{aligned} \quad (4.32)$$

Because of the cylindrical symmetry, the problem may be conveniently studied representing the incident fields in cylindrical harmonic functions through a Fourier series expansion of the fields in (4.32). In particular, the components of the fields tangential to the cylindrical surfaces can be expressed as

$$\begin{aligned} E_z^{\text{inc}}(\rho, \phi) &= E_0^{\text{inc}} \sum_{n=-\infty}^{\infty} j^{-n} J_n(k_0 \rho) e^{jn\phi}, \\ H_\phi^{\text{inc}}(\rho, \phi) &= \frac{1}{j\omega\mu_0} \frac{\partial E_z^{\text{inc}}}{\partial \rho} = -j \frac{E_0^{\text{inc}}}{\eta_0} \sum_{n=-\infty}^{\infty} j^{-n} J'_n(k_0 \rho) e^{jn\phi}, \end{aligned} \quad (4.33)$$

where $J_n(\cdot)$ represents the n th-order Bessel function of the first kind and the prime symbol means a derivative with respect to the argument, meaning $J'_n(\xi) = dJ_n(\xi)/d\xi$.

An analogous representation in terms of cylindrical harmonic functions may be considered for the scattered- and transmitted-field components in all the three regions. As is well known, cylindrical wave functions are solutions of the three-dimensional (3D) homogeneous Helmholtz equation in cylindrical coordinates [2]. In particular, the single cylindrical wave $\Psi^C(\rho, \phi, z)$ takes the form $\Psi^C(\rho, \phi, z) = P(\rho)\Phi(\phi)Z(z)$, where

$$\begin{aligned} Z(z) &= a_z \cos(k_z z) + b_z \sin(k_z z), \\ \Phi(\phi) &= a_\phi \cos(k_\phi \phi) + b_\phi \sin(k_\phi \phi), \\ P(\rho) &= a_\rho C_{k_\phi}^{(1)}(k\rho) + b_\rho C_{k_\phi}^{(2)}(k\rho). \end{aligned} \quad (4.34)$$

The functions $C_v^{(1)}$ and $C_v^{(2)}$ in (4.34) are any two independent cylindrical Bessel functions chosen among J_v , Y_v , $H_v^{(1)}$, and $H_v^{(2)}$. In bounded regions, oscillatory Bessel functions of the first and second kind (J_v and Y_v , respectively) are usually adopted to represent standing waves, whereas in unbounded regions, the Hankel functions of the first and second kind ($H_v^{(1)}$ and $H_v^{(2)}$) are used to represent inward and outward propagating waves, respectively. For z -independent problems (i.e., $k_z = 0$), it is assumed $Z(z) = 1$ and the cylindrical wave is simply $\Psi^C(\rho, \phi) = P(\rho)\Phi(\phi)$.

For the considered problem of Figure 4.8, periodicity in ϕ enforces k_ϕ to be an integer, meaning $k_\phi = n$. The reflected field in region 3, being an outward wave, requires the representation of its radial dependence through Hankel functions of the second kind $H_n^{(2)}$. The transmitted field inside the shield (region 2) will be instead represented in terms of both Bessel functions of the first and second kind J_n and Y_n , whereas the transmitted field inside the region 1 will be expressed in terms of the only J_n functions (since all the other functions are singular at the origin). It is immediate to see that it is not possible to satisfy the boundary conditions (i.e., continuity of the components of the fields tangential to the shield-air interfaces) with a single cylindrical wave function (each cylindrical wave function in fact cannot match the more complicated dependence on the spatial coordinates of the fields in (4.33)). However, since the set of cylindrical functions is complete, an infinite linear combination of Ψ^C functions with unknown coefficients can be used to represent the reflected and transmitted fields in regions 1, 2, and 3. After recognizing that the series of sinusoidal functions is a Fourier series, one can rewrite the components of the fields tangential to the cylindrical surfaces as

$$\begin{aligned} E_z^{(1)}(\rho, \phi) &= \sum_{n=-\infty}^{\infty} a_n j^{-n} J_n(k_0 \rho) e^{jn\phi}, \\ H_\phi^{(1)}(\rho, \phi) &= \frac{1}{j\omega\mu_0} \frac{\partial E_z^{(1)}}{\partial \rho} = -j \frac{1}{\eta_0} \sum_{n=-\infty}^{\infty} a_n j^{-n} J'_n(k_0 \rho) e^{jn\phi}, \end{aligned} \quad (4.35a)$$

$$E_z^{(2)}(\rho, \phi) = \sum_{n=-\infty}^{\infty} j^{-n} [b_n J_n(k\rho) + c_n Y_n(k\rho)] e^{jn\phi}, \quad (4.35b)$$

$$H_\phi^{(2)}(\rho, \phi) = -j \frac{1}{\eta} \sum_{n=-\infty}^{\infty} j^{-n} [b_n J'_n(k\rho) + c_n Y'_n(k\rho)] e^{jn\phi},$$

$$E_z^{(3)}(\rho, \phi) = \sum_{n=-\infty}^{\infty} j^{-n} [E_0^{\text{inc}} J_n(k_0\rho) + d_n H_n^{(2)}(k_0\rho)] e^{jn\phi}, \quad (4.35c)$$

$$H_\phi^{(3)}(\rho, \phi) = -j \frac{1}{\eta_0} \sum_{n=-\infty}^{\infty} j^{-n} [E_0^{\text{inc}} J'_n(k_0\rho) + d_n H_n^{(2)'}(k_0\rho)] e^{jn\phi},$$

where the prime symbol means again a derivative with respect to the argument $C'_n(\xi) = dC_n(\xi)/d\xi$. The unknown coefficients a_n , b_n , c_n , d_n , can be determined by enforcing the continuity of the tangential components of the electric and magnetic field at the shield-air interfaces (i.e., $\rho = \rho_1$ and $\rho = \rho_2$). By using the orthogonality relation among exponential functions, the following linear system is obtained:

$$\begin{bmatrix} J_n(k_0\rho_1) & -J_n(k\rho_1) & -Y_n(k\rho_1) & 0 \\ \frac{1}{\eta_0} J'_n(k_0\rho_1) & -\frac{1}{\eta} J'_n(k\rho_1) & -\frac{1}{\eta} Y'_n(k\rho_1) & 0 \\ 0 & J_n(k\rho_2) & Y_n(k\rho_2) & -H_n^{(2)}(k_0\rho_2) \\ 0 & \frac{1}{\eta} J'_n(k\rho_2) & \frac{1}{\eta} Y'_n(k\rho_2) & -\frac{1}{\eta_0} H_n^{(2)'}(k_0\rho_2) \end{bmatrix} \begin{bmatrix} a_n \\ b_n \\ c_n \\ d_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ E_0^{\text{inc}} J_n(k_0\rho_2) \\ \frac{E_0^{\text{inc}}}{\eta_0} J'_n(k_0\rho_2) \end{bmatrix}. \quad (4.36)$$

Once the system (4.36) is solved for each n , (4.35) gives the fields everywhere in space. However, in order to obtain the value of the *electric* SE on the cylinder axis, $\rho = 0$, only one coefficient is required (though this is not always the point of maximum field). For $\rho = 0$, all the functions in (4.35a) but the one corresponding to the coefficient a_0 are in fact identically zero. Therefore the total electric field on the cylinder axis in the presence of the shield E_z^S is $E_z^S(0, \phi) = a_0$; on the other hand, the electric field in the absence of the screen E_z^0 is simply the incident field, meaning $E_z^0(0, \phi) = E_0^{\text{inc}}$. The electric SE on the cylinder axis is thus

$$\text{SE}_{E(\text{dB})} = 20 \log \frac{|\mathbf{E}^0|}{|\mathbf{E}^S|} = 20 \log \left| \frac{E_z^0}{E_z^S} \right| = 20 \log \left| \frac{E_0^{\text{inc}}}{a_0} \right|, \quad (4.37)$$

where from (4.36),

$$a_0 = \frac{4jE_0^{\text{inc}}}{(\pi k_0 \eta_0)^2 \mu_r \rho_1 \rho_2} \cdot \begin{vmatrix} J_0(k_0\rho_1) & -J_0(k\rho_1) & -Y_0(k\rho_1) & 0 \\ \frac{1}{\eta_0} J'_0(k_0\rho_1) & -\frac{1}{\eta} J'_0(k\rho_1) & -\frac{1}{\eta} Y'_0(k\rho_1) & 0 \\ 0 & J_0(k\rho_2) & Y_0(k\rho_2) & -H_0^{(2)}(k_0\rho_2) \\ 0 & \frac{1}{\eta} J'_0(k\rho_2) & \frac{1}{\eta} Y'_0(k\rho_2) & -\frac{1}{\eta_0} H_0^{(2)'}(k_0\rho_2) \end{vmatrix}. \quad (4.38)$$

In order to obtain the *magnetic* SE, also the radial component of the magnetic field is needed. In particular, it simply results

$$H_\rho^{(1)}(\rho, \phi) = \frac{1}{j\omega\mu_0\rho} \frac{\partial E_z^{(1)}}{\partial \phi} = -\frac{1}{\eta_0} \sum_{n=-\infty}^{\infty} na_n j^{-n} \frac{J_n(k_0\rho)}{k_0\rho} e^{jn\phi}. \quad (4.39)$$

From (4.35a) and (4.36) and by using the properties of Bessel functions, it can thus be seen that $|\mathbf{H}^{(1)}(0, \phi)| = |a_1|/\eta_0$ (in particular, from $C_{-n}(\xi) = (-1)^n C_n(\xi)$, it follows that $a_{-n} = a_n$). Therefore, by taking into account that $|\mathbf{H}^{\text{inc}}(0, \phi)| = |E_0^{\text{inc}}/\eta_0|$, one can obtain the magnetic SE as

$$\text{SE}_{H(\text{dB})} = 20 \log \frac{|\mathbf{H}^0|}{|\mathbf{H}^S|} = 20 \log \left| \frac{E_0^{\text{inc}}}{a_1} \right|. \quad (4.40)$$

It should be noted that in this cylindrical configuration, contrary to what happens for uniform plane waves incident on a planar shield, the electric and magnetic SE are in general dramatically different.

The case of TE_z normal incidence can be studied in a similar way starting from

$$\begin{aligned} \mathbf{H}^{\text{inc}}(x, y, z) &= H_z^{\text{inc}}(x) \mathbf{u}_z = H_0^{\text{inc}} e^{-jk_0 x} \mathbf{u}_z, \\ \mathbf{E}^{\text{inc}}(x, y, z) &= E_y^{\text{inc}}(x) \mathbf{u}_y = E_0^{\text{inc}} e^{-jk_0 x} \mathbf{u}_y = \eta_0 H_0^{\text{inc}} e^{-jk_0 x} \mathbf{u}_y. \end{aligned} \quad (4.41)$$

In a still simpler way one can apply duality to obtain

$$\text{SE}_{E(\text{dB})} = 20 \log \frac{|\mathbf{E}^0|}{|\mathbf{E}^S|} = 20 \log \left| \frac{H_0^{\text{inc}}}{A_1} \right| \quad (4.42)$$

and

$$\text{SE}_{H(\text{dB})} = 20 \log \frac{|\mathbf{H}^0|}{|\mathbf{H}^S|} = 20 \log \left| \frac{H_z^0}{H_z^S} \right| = 20 \log \left| \frac{H_0^{\text{inc}}}{A_0} \right|, \quad (4.43)$$

where the unknown coefficients can be obtained by solving the linear system

$$\begin{bmatrix} J_n(k_0\rho_1) & -J_n(k\rho_1) & -Y_n(k\rho_1) & 0 \\ \eta_0 J'_n(k_0\rho_1) & -\eta J'_n(k\rho_1) & -\eta Y'_n(k\rho_1) & 0 \\ 0 & J_n(k\rho_2) & Y_n(k\rho_2) & -H_n^{(2)}(k_0\rho_2) \\ 0 & \eta J'_n(k\rho_2) & \eta Y'_n(k\rho_2) & -\eta_0 H_n^{(2)'}(k_0\rho_2) \end{bmatrix} \begin{bmatrix} A_n \\ B_n \\ C_n \\ D_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ H_0^{\text{inc}} J_n(k_0\rho_2) \\ \eta_0 H_0^{\text{inc}} J'_n(k_0\rho_2) \end{bmatrix}. \quad (4.44)$$

Figure 4.9 shows the SE versus frequency of a glass cylindrical surface with $\varepsilon_r = 3.8$, $\rho_1 = 10$ cm, and $\rho_2 = 10.1$ cm. It is possible to note that for both the TM_z

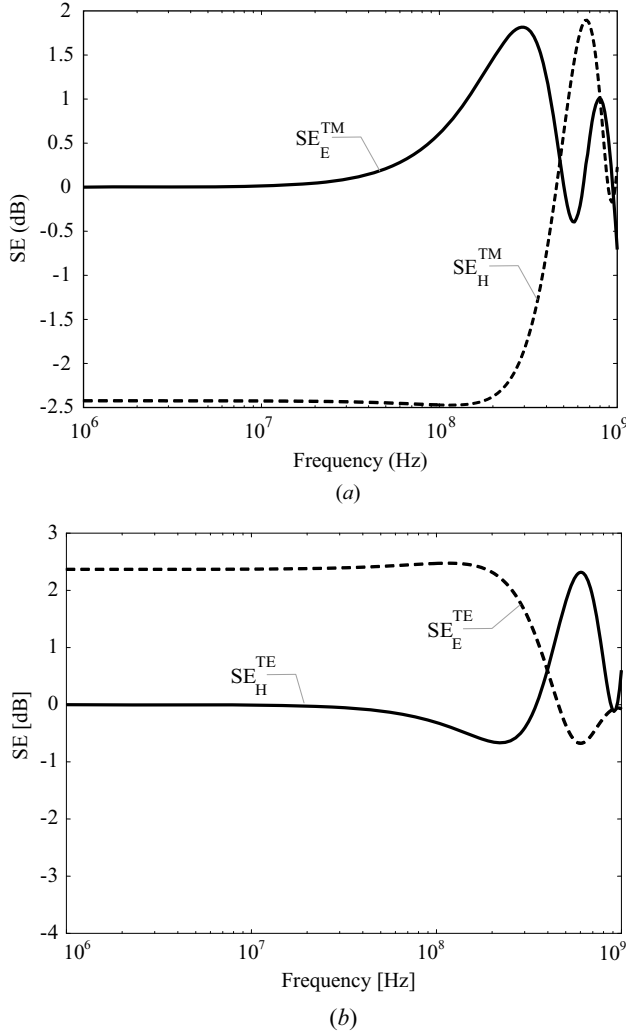


FIGURE 4.9 SE of a glass cylindrical surface with $\varepsilon_r = 3.8$, $\rho_1 = 10$ cm, and $\rho_2 = 10.1$ cm under TM_z and TE_z normal incidence.

and TE_z normal incidence, the electric and magnetic SE are very different. In addition negative values of SE can be observed.

The problem of EM plane-wave transmission through a hollow metallic cylinder has been studied in [6], where the shield thickness was assumed electrically small, and next, for arbitrary thickness values, in [7].

A numerical study of the normal plane-wave incidence on a circular cylindrical shield consisting of a single dielectric layer is reported in [8], and results for multilayer circular cylindrical structures appear in [9]. The oblique plane-wave

incidence on a single dielectric cylinder (which couples TM_z and TE_z fields) was first studied in [10], and the cylindrical shell was considered in [11]. The case of anisotropic multilayer cylinders is studied in [12] and [13], while in [14] both anisotropy and nonlinearity are taken into account. An extension of the formalism to deal with elliptical shields (which involves the use of Mathieu functions) is reported in [15]. An interesting optimization procedure, based on a genetic-algorithm approach, has been developed in [16] in order to determine the optimal thicknesses and type of materials of a multilayer cylindrical shell to obtain a given SE in a particular range of frequencies and of incidence angles.

Finally, it is worth noting that cylindrical shields may act (especially in the high-frequency region) as cavity resonators, and at the resonant frequencies the shielding performance can dramatically get worse. A study of such resonance effects has been carried out in [17] for the general case of oblique incidence.

4.4 PLANE WAVES AGAINST SPHERICAL SHIELDS

The scattering from a sphere is one of the few 3D problems that can be solved in a closed form, and also the simplest. In fact, the problem of plane-wave incidence on a multilayered spherical shield is formally equivalent to the one involving normal incidence on a multilayer cylindrical shield, except that spherical-wave expansions are used instead of cylindrical-wave expansions. *Spherical-wave functions* are solutions of the 3D homogeneous Helmholtz equation in spherical coordinates [2]. In particular, the single spherical wave $\Psi^S(r, \phi, \theta)$ takes the form $\Psi^S(r, \phi, \theta) = R(r)\Phi(\phi)\Theta(\theta)$ where

$$\begin{aligned}\Phi(\phi) &= a_\phi \cos(\mu\phi) + b_\phi \sin(\mu\phi), \\ \Theta(\theta) &= a_\theta P_\nu^\mu(\cos\theta) + b_\theta Q_\nu^\mu(\cos\theta), \\ R(r) &= a_r S_\nu^{(1)}(kr) + b_r S_\nu^{(2)}(kr).\end{aligned}\tag{4.45}$$

The functions P_ν^μ and Q_ν^μ (with $\mu^2 = \nu(\nu+1)$) are the associated Legendre functions of the first and second kinds, respectively, and $S_\nu^{(1)}$ and $S_\nu^{(2)}$ are two independent *spherical* Bessel functions chosen among j_ν , y_ν , $h_\nu^{(1)}$, and $h_\nu^{(2)}$. It should be noted that in problems containing the z axis, the b_θ coefficient must be considered zero in order to avoid singularities; moreover ν and μ are restricted to be integers ($\nu = n$ and $\mu = m$). The product of the functions Φ and Θ are usually called *spherical harmonics* ($Y_{mn}(\phi, \theta) = \Phi(\phi)\Theta(\theta)$). In problems involving spherical waves, the TE-TM decomposition of a given EM field in spherical coordinates is usually performed with respect to the r direction.

Several papers in the literature analyze scattering from a multilayered sphere as shown in Figure 4.10a. Aden and Kerker were first to investigate the incidence of a uniform plane wave on a concentric spherical shell [18], and shortly after, in [19], approximate results were obtained in the limit of very thin shells. Multilayered spherical shields were studied in [20–22], and the possible anisotropy of the

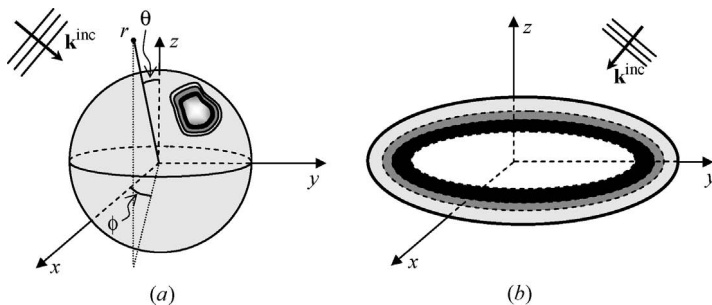


FIGURE 4.10 Plane-wave incidence on a multilayer spherical shield (a) and on a multilayer spheroidal shield (b).

materials was taken into account in [23] for a single sphere and in [24] for a multilayered spherical structure. Resonance effects in a spherical shell were studied in [25], and, more recently, in [26] the use of lossy dielectrics was suggested to mitigate such undesired effects, determining (for a double-layered spherical shield) an approximate value for the optimum conductivity of a metal layer that minimizes the Q factor at the fundamental resonant frequency of the structure.

Other 3D shielding problems solvable in closed form involve shields of ellipsoidal shape, as shown in Figure 4.10b. The mathematical formalism requires the use of spheroidal wavefunctions [27], and analyses of different configurations (involving a single spheroid, a spheroidal shell, and a multilayer spheroid) can be found in [28–30].

4.5 LIMITS TO THE EXTENSION OF THE TL ANALOGY TO NEAR-FIELD SOURCES

Since the analytical solution of the field equations is complicated, even for simple shield configurations, the TL approach has been widely used for evaluating the performances of shields also against near-field sources, because of its inherent formal simplicity. However, impinging fields are different from plane waves, and so require that the TL analogy be carefully used to avoid incorrect results.

Generally, when a planar screen is placed in the near-field region of a finite source, the incident EM field produced by the source can be represented by an integral superposition of plane waves having all the possible directions of propagation and polarizations. In principle, one could apply the TL approach to each of these elemental plane waves, and then reconstruct the field beyond the screen by summing up the contributions of all the transmitted elemental plane waves. However, the TL parameters would be different for different plane waves and hence a unique TL cannot be defined for the total field.

Despite these considerations, an approximate TL approach does exist for the evaluation of the SE of an infinite highly conductive planar screen in the presence of

a near-field source, and it has been widely used through the years [31–32]. As shown in Section 4.2.2, when the planar screen is highly conductive, the EM field inside it is (with very good approximation) that of a uniform plane wave propagating in a direction normal to the screen, independently of the incident field. Therefore a TL in the direction normal to the screen can be introduced to model the field propagation inside it.

However, it is not clear how to establish a TL model for the field outside the screen, since, as explained above, each elemental plane wave in the spectrum of the incident field is modeled by a different TL. In order to overcome this difficulty, it may be noted that for a single plane wave, the characteristic impedance of its equivalent TL coincides with its *wave impedance* in the TL direction. According to [33], the wave impedance along an arbitrary direction $\mathbf{u}$ is defined as the ratio of the transverse (with respect to $\mathbf{u}$) electric-field component to the mutually perpendicular transverse magnetic-field component. Therefore the idea is to use the wave impedances of the incident field at the screen interfaces as the characteristic impedances of the TLs modeling the field outside the screen.

It should be noted that the wave impedances of the incident field actually depend on the observation point on the screen interface. For instance, let us consider an infinitesimal electric or magnetic dipole, which can be used as a first approximation to model finite sources of small dimensions. With respect to the arbitrarily oriented reference system (x', y', z') , the electric- and magnetic-field components produced by an infinitesimal electric dipole (ED) of moment $I\mathbf{u}_{z'}$ can be expressed in spherical coordinates as

$$\begin{aligned} E_r(r', \theta') &= \eta_0 \frac{I}{2\pi} \cos \theta' \left[\frac{1}{r'^2} + \frac{1}{jk(r')^3} \right] e^{-jkr'}, \\ E_\theta(r', \theta') &= \eta_0 \frac{I}{4\pi} \sin \theta' \left[\frac{jk}{r'} + \frac{1}{(r')^2} + \frac{1}{jk(r')^3} \right] e^{-jkr'}, \\ H_\phi(r', \theta') &= \frac{I}{4\pi} \sin \theta' \left[\frac{jk}{r'} + \frac{1}{(r')^2} \right] e^{-jkr'}, \end{aligned} \quad (4.46)$$

while an infinitesimal magnetic dipole (MD) of moment $IS\mathbf{u}_{z'}$ produces a field

$$\begin{aligned} E_\phi(r', \theta') &= \frac{j\omega\mu_0 IS}{4\pi} \sin \theta' \left[\frac{jk}{r'} + \frac{1}{(r')^2} \right] e^{-jkr'}, \\ H_r(r', \theta') &= \frac{j\omega\mu_0 IS}{2\pi\eta_0} \cos \theta' \left[\frac{1}{(r')^2} + \frac{1}{jk(r')^3} \right] e^{-jkr'}, \\ H_\theta(r', \theta') &= \frac{j\omega\mu_0 IS}{4\pi\eta_0} \sin \theta' \left[\frac{jk}{r'} + \frac{1}{(r')^2} + \frac{1}{jk(r')^3} \right] e^{-jkr'}. \end{aligned} \quad (4.47)$$

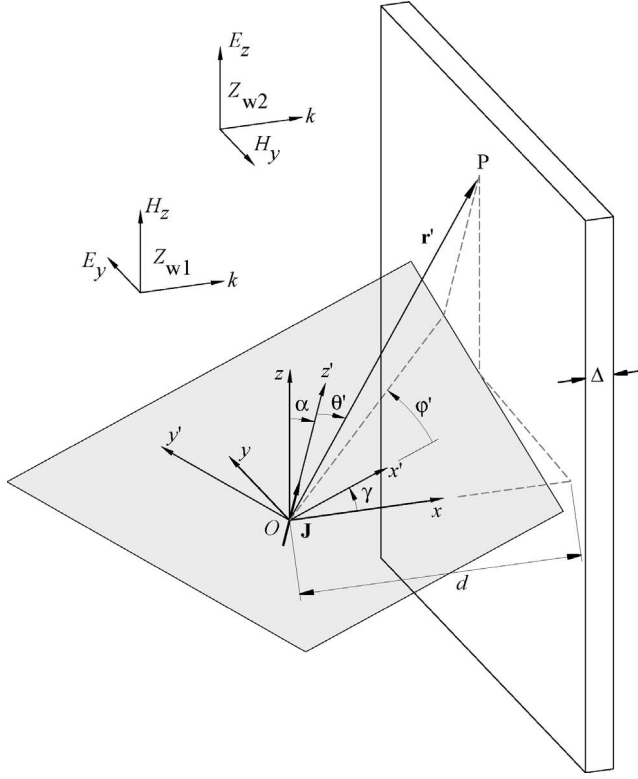


FIGURE 4.11 Infinitesimal dipole source arbitrarily oriented illuminating an infinite planar shield.

With respect to the reference system (x, y, z) shown in Figure 4.11, two wave impedances Z_{w1} and Z_{w2} can be defined as

$$Z_{w1} = + \frac{E_y^{\text{inc}}}{H_z^{\text{inc}}}, \quad (4.48a)$$

$$Z_{w2} = - \frac{E_z^{\text{inc}}}{H_y^{\text{inc}}}. \quad (4.48b)$$

It is possible to move from the unprimed reference system to the primed one by a rotation around the z axis of an angle γ and then around the new rotated x axis of an angle α . The wave impedances are then given by

$$Z_{w1}^{\text{ED}}(r, \phi, \theta) = + \frac{s_{22}E_r(r', \theta') + s_{23}E_\theta(r', \theta')}{s_{21}H_\phi(r', \theta')}, \quad (4.49a)$$

$$Z_{w2}^{\text{ED}}(r, \phi, \theta) = - \frac{s_{12}E_r(r', \theta') + s_{13}E_\theta(r', \theta')}{s_{11}H_\phi(r', \theta')},$$

$$Z_{w1}^{\text{MD}}(r, \phi, \theta) = + \frac{s_{11}E_\phi(r', \theta')}{s_{12}H_r(r', \theta') + s_{13}H_\theta(r', \theta')}, \quad (4.49b)$$

$$Z_{w2}^{\text{MD}}(r, \phi, \theta) = - \frac{s_{21}E_\phi(r', \theta')}{s_{22}H_r(r', \theta') + s_{23}H_\theta(r', \theta')},$$

for an electric and a magnetic dipole, respectively. Here

$$\begin{aligned} s_{11} &= \cos \alpha \cos \gamma \cos \phi' - \sin \gamma \sin \phi', \\ s_{12} &= \cos \alpha \cos \theta' + \sin \alpha \sin \theta' \sin \phi', \\ s_{13} &= \cos \alpha \sin \theta - \sin \alpha \cos \theta' \sin \phi', \\ s_{21} &= \sin \alpha \cos \phi', \\ s_{22} &= -\sin \alpha \cos \gamma \cos \theta' + \sin \gamma \sin \theta' \cos \phi' + \cos \alpha \cos \gamma \sin \theta' \sin \phi', \\ s_{23} &= \sin \gamma \cos \theta' \cos \phi + \sin \alpha \cos \gamma \sin \theta' + \cos \alpha \cos \gamma \cos \theta' \sin \phi'. \end{aligned} \quad (4.50)$$

From (4.49) and (4.50), it can be seen that the wave impedances strongly depend on the observation point. In the TL analogy, the SE is the sum of three terms: the reflection-loss term R , the absorption-loss term A , and the multiple-reflection-loss term M . However, both R and M depend on the mismatch between the wave impedance and the intrinsic impedance of the shield medium η . It is clear that the SE of the screen cannot be fully evaluated if only the shield characteristics are known (i.e., its geometrical and physical properties), since the SE strictly depends also on the size, location, and orientation of the source with respect to the shield. It should be noted that the source identification is a necessary prerequisite to any shielding analysis. A first fundamental step in the source-identification process when dealing with near fields is the distinction between high- and low-impedance field sources. High-impedance sources produce a near field dominated by the electric field $\mathbf{E}$ (i.e., $|Z_{w1,2}| \gg \eta_0$), while low-impedance field sources are characterized by a near field dominated by the magnetic field $\mathbf{H}$ (i.e., $|Z_{w1,2}| \ll \eta_0$).

In practice, when the TL analogy is applied to evaluate the SE of an infinite planar conducting screen, the impedances of the impinging field are computed along the axis drawn from the source orthogonally to the shield (x axis in Figure 4.11). Two significant configurations are those in which a dipole is normal to the shield (i.e., $\gamma = 0$ and $\alpha = \pi/2$) or parallel to the shield (i.e., $\gamma = 0$ and $\alpha = 0$). From equations (4.46) through (4.50), by assuming $\theta' = 0$ for orthogonal dipoles ($\perp$) and $\theta' = \pi/2$ for parallel dipoles ($\parallel$), respectively, the following expressions for the wave impedances can be obtained:

$$Z_{w2}^{\text{ED}\perp}(x, \omega) = \eta_0 \frac{\xi^2 + \xi + 1}{\xi^2 + \xi}, \quad (4.51a)$$

$$Z_{w1,2}^{\text{MD}\perp}(x, \omega) = \eta_0 \frac{\xi^2 + \xi}{\xi^2 + 3\xi + 3}, \quad (4.51b)$$

$$Z_{w1,2}^{\text{ED}\parallel}(x, \omega) = \eta_0 \frac{\xi^2 + 3\xi + 3}{\xi^2 + \xi}, \quad (4.51c)$$

$$Z_{w1}^{\text{MD}\parallel}(x, \omega) = \eta_0 \frac{\xi^2 + \xi}{\xi^2 + \xi + 1}, \quad (4.51d)$$

where $\xi = jk_0 x$. It should be noted that the wave impedances are both defined only when the electric dipole is parallel to the shield or the magnetic dipole is perpendicular to the screen.

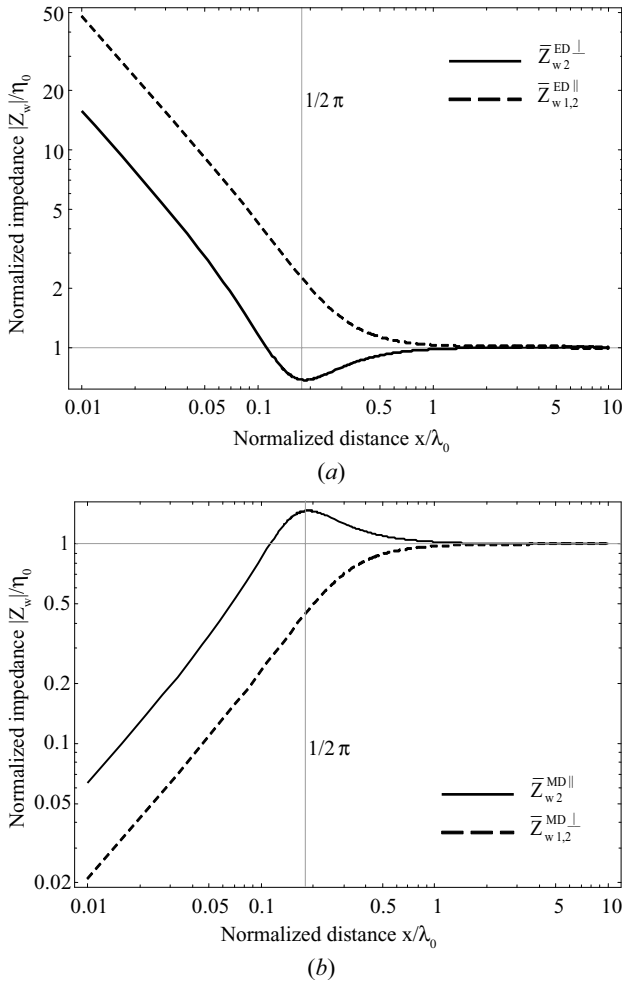


FIGURE 4.12 Magnitude of the normalized wave impedances as functions of the normalized distance $\bar{x} = x/\lambda_0$. Electric dipole (a) and magnetic dipole (b).

The behavior of the wave impedances is illustrated in Figure 4.12, where the normalized wave impedances $|Z_{w1,2}^{ED/MD}| = |Z_{w1,2}^{ED/MD}|/\eta_0$ are reported as functions of the normalized distance $\bar{x} = x/\lambda_0$. For normalized distances smaller than $1/(2\pi)$ (i.e., in the near-field region), the wave impedances of the field produced by the electric dipole are larger than the free-space impedance η_0 , while those corresponding to the magnetic dipole are smaller. It can be concluded that the electric dipole is a high-impedance field source, while the magnetic dipole is a low-impedance field source.

To illustrate how the TL approach behaves when it is extended to study near-field sources, the typical values of the intrinsic impedance of shielding materials are compared with the wave impedances of the impinging field produced by the source. From Figure 4.3 it can be seen that the intrinsic impedance of shielding materials is much smaller than the free-space impedance, also when the frequency is increased up to 1 GHz. In the case of high-impedance field sources, the ratio $\zeta = |Z_{w1,2}^{ED/MD}|/\eta$ is so large that the reflection-loss term R always gives a remarkable contribution to the overall SE. Furthermore this type of sources usually works at high frequencies (VHF, UHF, and SHF) where the screen is electrically thick and the absorption-loss term A gives a significant contribution as well. The values of the SE are so large that the approximations due to the extension of the TL analogy to near-field sources and to the small-dipole approximation of the actual sources do not affect the overall accuracy of the results. The TL approach is particularly convenient here because it provides accurate results, even if the actual source is modeled as a small dipole. The effect of the main quantities on the SE is shown in Figure 4.13. A copper planar screen ($\sigma = 11.8 \cdot 10^6$ S/m, $d = 0.5$ mm) of infinite extent is assumed to be placed 30 cm far from an electric dipole parallel to the shield. In the low-frequency range the reflection-loss term R is the dominant

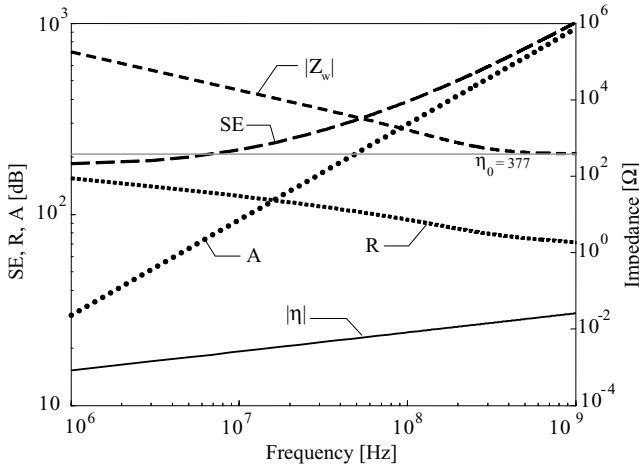


FIGURE 4.13 Magnitude of the contributions to the SE of an infinite copper planar screen of thickness $d = 0.5$ mm illuminated by a parallel electric dipole as a function of frequency.

contribution to the SE, due to the several orders of mismatch between the incident-wave impedances $|Z_{w1,2}^{ED}|$ and the intrinsic characteristic impedance of copper $|\eta|$ (the multiple-reflection-loss term M is practically zero). As the frequency is increased, the screen becomes more and more electrically distant from the source; the values $|Z_{w1,2}^{ED}|$ of the wave impedances decrease toward that of the free-space impedance of a normally incident plane wave η_0 , the characteristic impedance $|\eta|$ of copper increases, and the contribution of R to the SE reduces while the contribution of A increases. The SE always assumes large values, varying from 150 to 1000 dB. In this range of values, the errors introduced by the TL analogy are negligible.

On the other hand, in the case of low-impedance field sources, the TL analogy needs to be applied with several attentions to avoid incorrect results. Low-impedance field sources (e.g., loops or straight conductors carrying an electric current) are mainly source of magnetic field in the low-frequency range (e.g., ULF and VLF). At low frequencies the conducting shield is electrically small ($d \ll \lambda_0$), and the contribution of the absorption-loss term A is almost negligible. For the materials shown in Figure 4.3, the magnitude of the wave impedances is from one to three orders of magnitude larger than that of their intrinsic impedance $|\eta|$: the reflection-loss term is the dominant contribution to the SE, although its values are much smaller than those attained in the case of high-impedance field sources. Moreover the multiple-reflection-loss term M can assume negative values, further reducing the SE values. This behavior is illustrated in Figure 4.14, where a copper planar screen ($\sigma = 11.8 \cdot 10^6$ S/m, $d = 0.5$ mm) of infinite extent is assumed to be placed 30 cm far from a magnetic dipole parallel to the screen. Below 200 Hz, the SE is practically zero and the R and M terms are almost equal but opposite in sign. From 200 up to 200 kHz,

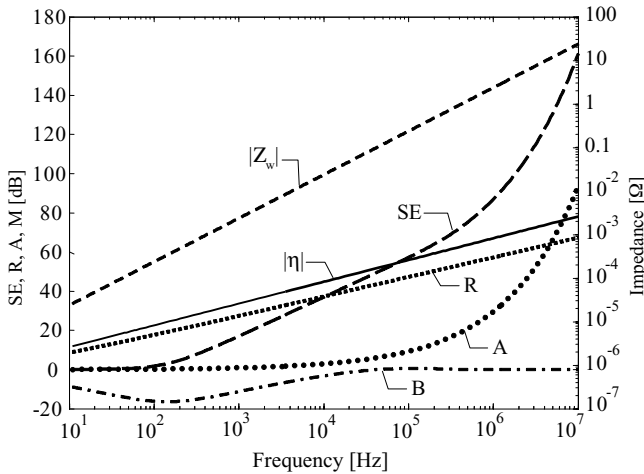


FIGURE 4.14 Magnitude of the contributions to the SE of an infinite copper planar screen illuminated by a parallel magnetic dipole as a function of frequency.

the main contribution to the SE is mainly due to the reflection-loss term R , since the mismatch between the wave impedances and the copper intrinsic impedance increases. Above 200 kHz, the absorption-loss term A starts to contribute to the overall SE while the contribution of the M term drops to zero.

A comparison of Figures 4.13 and 4.14 shows that a difference of about one order of magnitude exists between the SE of the screen when it is illuminated by the high-impedance field source and by the low-impedance field source. From a physical point of view, the main difference is the role played by the eddy-current cancellation mechanism (see Appendix B), which is practically negligible at very low frequencies, and the different weight of the reflection- and multiple-reflection-loss terms. For low-impedance field sources these two terms are the dominant contributions to the SE. Hence any small error in the computation of these terms deeply affects the accuracy of the final results provided by the TL analogy. For this reason it is important to compute the wave impedances of the field produced by actual sources as accurately as possible.

In Appendix B some canonical problems of VLF magnetic shielding are revised and the analytical solutions are provided. These problems deal with typical sources of practical interest, as illustrated in Figure 4.15: the current loop placed in a plane normal to the shield, the current loop placed in a plane parallel

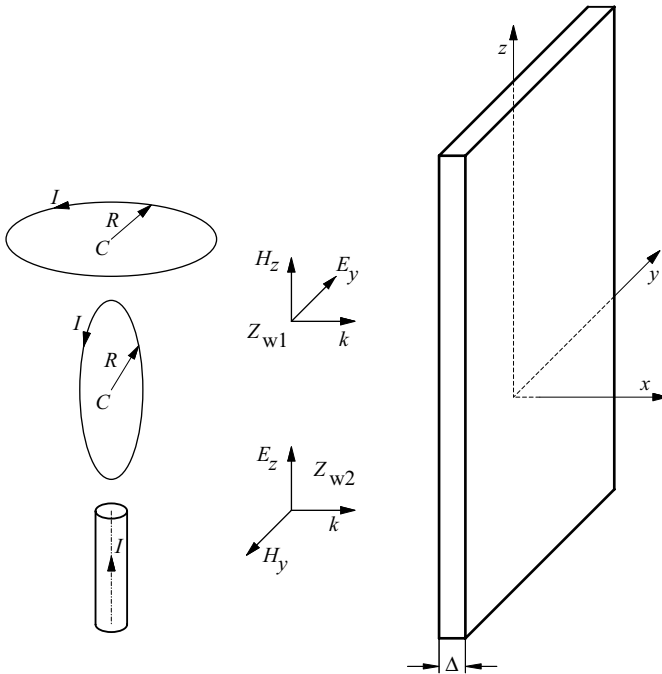


FIGURE 4.15 Basic low-frequency field sources.

to the shield, the straight wire conductor parallel to the shield. Comparisons between the results obtained through exact formulations and the approximate TL analogy are also reported, showing that the latter can provide very accurate results.

For these sources it is possible to compute the correct wave impedances along the normal to the shield (x axis) as

$$Z_{w1}^{\text{loop}\perp} = -j\omega\mu_0 \frac{\int_0^\infty [\int_0^\infty [I_1(\beta R)/\beta] e^{-\sqrt{\alpha^2+\beta^2}x} d\alpha] d\beta}{\int_0^\infty [\int_0^\infty [\beta I_1(\beta R)/\sqrt{\alpha^2+\beta^2}] e^{-\sqrt{\alpha^2+\beta^2}x} d\alpha] d\beta}, \quad (4.52a)$$

$$Z_{w1,2}^{\text{loop}\parallel} = -j\omega\mu_0 \frac{\int_0^\infty \left(\nu/\sqrt{\nu^2 - k_0^2} \right) J_1(\nu R) e^{-\sqrt{\nu^2 - k_0^2}x} d\nu}{\int_0^\infty \nu J_1(\nu R) e^{-\sqrt{\nu^2 - k_0^2}x} d\nu}, \quad (4.52b)$$

$$Z_{w2}^{\text{wire}\parallel} = \eta_0 \frac{K_0(jk_0x)}{K_1(jk_0x)}, \quad (4.52c)$$

where $I_1(\cdot)$ is the first-order modified Bessel functions of the first kind while $K_0(\cdot)$ and $K_1(\cdot)$ are the zero- and first-order modified Bessel functions of the second kind, respectively.

In Figure 4.16, a comparison among these wave impedances (radius of the loop $R = 30$ cm) and the wave impedances of an infinitesimal magnetic dipole for a shield-to-source distance of 30 cm is shown. Even if the sources are all characterized by very low values of the wave impedances, some differences are noticeable. Hence, when a real source is replaced with a small magnetic dipole, the wave impedances

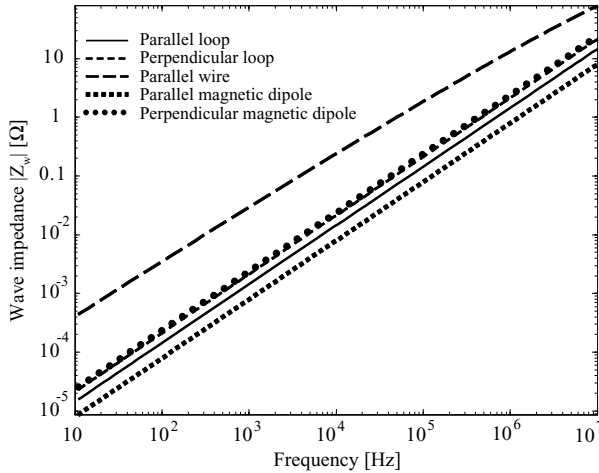


FIGURE 4.16 Comparison among the wave impedances of low-frequency field sources.

must be accurately evaluated since small differences can provide inaccurate results when the TL analogy is adopted.

In the low-frequency range, where the approach is more critical, the TL analogy has been widely tested with high-permeability and high-conductivity materials [1, 31, 32, 37], showing results in reasonable agreement with those obtained through measurements and those calculated with exact formulations [33–37].

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